

2020 AMC 10A Problem 3 - Solution

Review the full statement and step-by-step solution for 2020 AMC 10A Problem 3. Great practice for AMC 10, AMC 12, AIME, and other math contests

2020 AMC 10A Problem 3

Problem:

Assuming $a \neq 3$, $b \neq 4$, and $c \neq 5$, what is the value in simplest form of the following expression?

$$\frac{a-3}{5-c} \cdot \frac{b-4}{3-a} \cdot \frac{c-5}{4-b}$$

Answer Choices:

- A. -1
- B. 1
- C. $\frac{abc}{60}$
- D. $\frac{1}{abc} - \frac{1}{60}$
- E. $\frac{1}{60} - \frac{1}{abc}$

Solution:

If $x \neq y$, then

$$\frac{x - y}{y - x} = -1$$

We use this fact to simplify the original expression:

$$\begin{aligned} & \frac{a - 3}{5 - c} \cdot \frac{b - 4}{3 - a} \cdot \frac{c - 5}{4 - b} \\ &= \frac{a - 3}{3 - a} \cdot \frac{b - 4}{4 - b} \cdot \frac{c - 5}{5 - c} \\ &= -1 \cdot -1 \cdot -1 \\ &= \boxed{(A) - 1} \end{aligned}$$

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